Integrated Approach To Assess The Quality Of Water

C. Sudheer Kumar[1] [a], Suhana Anjum S K[1] [b], Mahammad Thanveer P[1] [c] , Pallavi C[1] [d]
[1] *Srinivasa Ramanujan Institute of Technology*
Department of Computer Science and Engineering,
Anantapur, Andhra Pradesh, 515001, India
sudheer.cse@srit.ac.in, 224g1a05A7@srit.ac.in, 224g1a0546@srit.ac.in, 224g1a0564@srit.ac.in

Keywords: River water quality, water pollution, object detection, deep learning, environmental monitoring, plastic pollution, eutrophication, automated assessment, computer vision, Flask framework.

Abstract: Water pollution can be a very serious issue to the ecosystem and health to people and hence proper monitoring is necessary. The proposed project facilitates a combined mechanism of evaluating the quality of river water by automated monitoring of pollutants and debris. The system is a deep learning-based detection model which is integrated with a web-based interface, which allows users to download images of river water and receive a systematic study of types of pollution, such as acid contamination, plastic waste, oil residues, eutrophication, and dead organisms, as well as objects, such as bottles, paper, metal, and other debris. The sophisticated model of object detection was trained on a varied annotated dataset, which displayed a high status of detection in the identification of common objects and the occurrence of problems in classification of objects that resemble each other visually like dead animals and background. Comparison based on normalized confusion matrices reveals that plastic, metal, and paper detection have a high level of performance, and false identifications are mainly in the complex categories. Automated notifications and organized data storage are also included in the system, which helps to monitor and document the environment on the fly. This project has effectively combined effective detection algorithms, user interaction modules, and alert, thus it offers a scaled and stable framework to monitor river water at all times. The approach can be applied to other water bodies and environmental monitoring application and it provides a proactive tool to manage and mitigate water pollution.

1 INTRODUCTION

Water resources are important in sustaining ecosystems, human health, agriculture and economic activities. Rivers are among the resources that are prone to pollution because of high rate of urbanization, industrial effluents, agricultural runoff and poor disposal of waste products. Constant inflow of chemical substances, organic waste, residual oil, and plastic debris into rivers has resulted in the extreme deterioration of the water quality and aquatic organisms. The pollution of rivers has thus emerged as a critical issue of concern on the environment that needs effective and reliable methods of assessment.

Traditional water quality assessment methods are mostly manual-based and lab-based. Although such techniques are able to give accurate measurements of chemicals and biological parameters, they tend to be laborious, costly, and unable to cover the spatial and temporal scale. Moreover, conventional methods cannot happen to detect the pollutants that can be seen in the surface like plastic waste, dead organisms, oil films, and floating debris, which are significant signs of environmental degradation. Consequently, automated and scalable solutions with the potential to supplement conventional measures of water quality are becoming increasingly demanded.

The recent breakthrough in the area of artificial intelligence and computer vision has presented new possibilities of environmental monitoring. The analysis which is based on images makes it possible to identify patterns of visible pollution in the water bodies even without physical contact and in a wider area because it is faster. Algorithms of object

[a] https://orcid.org/0000-0001-5019-3582

detection, especially, have shown good potential in object detection and classification of various objects in complex visual images. These methods can be greatly used to improve the effectiveness and uniformity of monitoring river pollution.

The proposed project introduces a unified system of analyzing river water quality based on deep learning-based object detection. The system is capable of detecting all types of pollution like acid pollution, eutrophication, oil spills, dead animals and plastic waste and general litter like bottles, metal, glass, paper, and leaves. Total pollutant classification of multiple pollutants in one image with the help of a single image is possible with a unified detection strategy based on YOLOv8, YOLOv9 and YOLOv10. It has a web-based implementation based on a Python backend and interactive front end where users may upload an image and get automated results of the analysis.

The proposed system allows combining object detection with the interaction of the user and automated reporting in order to consider a structured and efficient solution to the problem of river water quality assessment. The framework assists in systematic documentation and can be expanded to other water bodies, which will enhance better environmental monitoring and informative decision-making.

2 RELATED WORK

In the company of recent progress in the field of water quality assessment, it is possible to note a growing tendency to resort to automated and smart methods of monitoring and overcoming the constraints of the old ones. Computer vision and image-based strategies have also received attention due to their detection capabilities with visible pollutants in water bodies. It has been shown that deep learning models can be used to detect floating wastes and plastic containers in water bodies with high reliability, which can be compared to manual tools of inspection (A. Heller et al 2025). Such methods enhance the detection performance and permit extensive rivers and lakes monitoring.

The application of deep learning to contaminants and biological indicators detection in water is discussed in several pieces of work. Robotic solutions that can detect the presence of bacteria and the patterns of pollution based on visual data have demonstrated the encouraging results in terms of faster response and the accuracy of assessment (R. Saleem 2024). Detection technology has been developed to detect objects on water surfaces considering the problem of reflections, differing light, and the background which is not smooth (W. Zhu 2025). These are some of the improvements that make the detection systems of pollution stronger in the real world.

Studies have also focused on the identification of solid waste specifically the plastic debris that has become a significant menace to the aquatic ecosystem. The detection of plastic bottles and floating waste via deep learning has been proven to detect the surface-level waste, which shows the possibilities of vision-based structures in pollution control strategies (N. L. Kirana 2024). The methods of predictive modeling and neural network-based have also been used to complement water quality assessment by acquiring trends and patterns of pollution over time (M. A. Novianta et al 2024).

In addition to local image analysis, remote sensing and hyperspectral imaging have been coupled with deep learning to evaluate the water quality in a greater spatial scale. Such approaches allow retrieving the environmental indicators and the level of pollution based on satellite and aerial imagery, which helps to implement a complete monitoring solution (Z. Pang 2025),(H. Li et al 2024). Elucidable artificial intelligence has been presented as well to enhance the interpretability of deep learning models utilized in the evaluation of water quality to maintain transparency in decision-making processes (S. Alqadhi 2025).

It has been proved, based on the recent studies, that remote sensing data used together with deep learning can be effectively used to analyze pollution specific to a river and monitor the environment (P. Mathur 2025). Optical systems have also been used to estimate optical parameters of water and to aid in automated watchtower systems (M. Wilchek 2026). Other studies have investigated the hybrid object detection models and AI- controlled surveillance systems to increase the monitoring capabilities of the river (H. Retief 2026), (P. S. K, Rathik 2025). Specifically, designed waste detection systems that suit particular river settings only confirm the flexibility of the deep learning models to a variety of geographic settings (A. Surahmat 2025) Also, the monitoring and real- time prediction models through video have increased the potential of automated water assessment systems (J. Kim 2025), (S. Toumi et al 2024). The articles show that deep learning and computer vision have developed into essential technologies which scientists use to build sustainable water quality monitoring systems that operate with high efficiency.

3 PROPOSED METHODOLOGY

The proposed system methodology is aimed at the automated framework of river water quality evaluation by detecting pollution using images. The general idea is data preparation, object detection by means of deep learning, and integration of the system to allow a user and report. The framework will be such that it is accurate, scalable, and easy to use and provide systematic monitoring of river pollution.

A. Dataset Description and Preparation

The initial phase of the methodology is gathering a wide set of data that reflects different pollution situations of rivers. This dataset is composed of the civil images of river water with various forms of contamination including acid pollution, eutrophication, fish oil remnants, dead species, and plastic debris. Moreover, the typical floating and submerged items such as bottles, cardboard, glass, metal, paper, leaves, and other litter are also taken into consideration in order to increase the coverage of detection. The pictures are collected under different sources and under different environmental situations to ensure that there is variability in the lighting conditions, water clarity and the complexity of the backgrounds.

Model performance and consistency are improved through the process of preprocessing. The process includes three steps which are to scale images to a standard resolution and to normalize pixel values and to remove all defective images. The annotation process requires manual work as pollutants and objects must be identified through bounding box labels which helps to teach accurate localization. Data augmentation techniques including rotation and flipping and brightness adjustment work together to produce diverse data which reduces the risk of overfitting. The dataset is divided into three sets which include training and validation and testing to enable effective learning and performance assessment.

Figure 1 Dataset Image

Figure 2 Dataset Image

Figure 3 Dataset Image

B. Object Detection Model and Training

The main part of the suggested methodology will be the use of deep learning-based object detection to detect the pollutants and debris in the images of river water. The detection system employs YOLOv8, and YOLOv9, and YOLOv10 to allow the real-time and precise classification of several types of pollution in one image. These models are chosen due to their effectiveness in the processing of complicated visual images and their capability of identifying small and intersecting objects which are usual in the water setting.

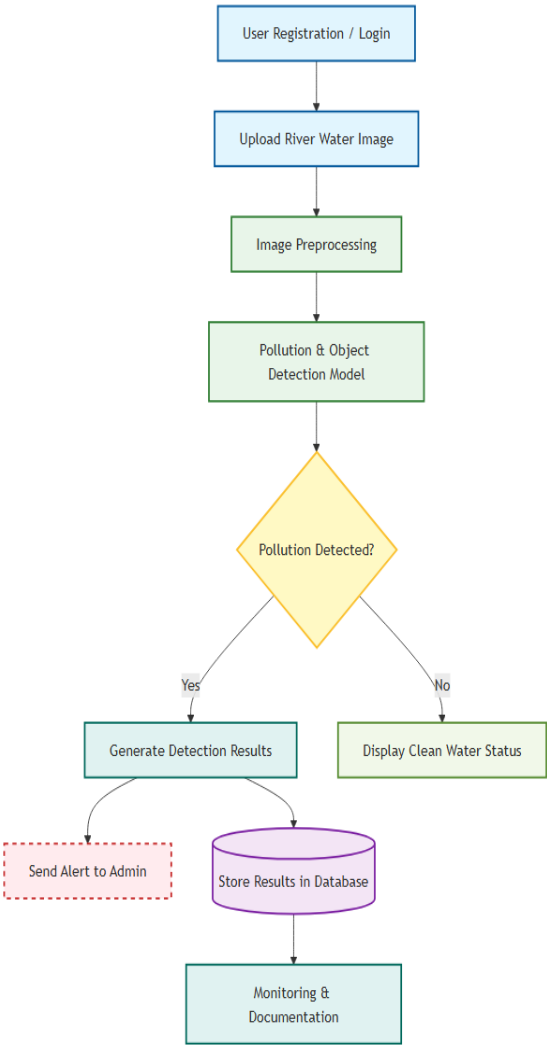

Figure 4 Proposed Methodology flow

The model training process uses the annotated dataset which enables the network to acquire spatial and contextual knowledge about different pollutants. The system achieves maximum detection accuracy through precise adjustment of three parameters which include learning rate, batch size and number of epochs. The training process uses precision, recall and mean average precision as performance metrics to assess model effectiveness. The testing process evaluates trained models using unfamiliar data to assess their ability to generalize and perform in various environmental settings.

C. System Integration and Deployment

The last phase will entail deploying the trained detection model into a web-based application to be deployed. This is accomplished by the implementation of a Python-based backend based on the Flask framework and the front end is created using HTML, CSS, and JavaScript to have an interactive interface. The users are able to create an account, log in, and post images of river water to be analyzed. After sending an image, the detection module takes it through, and the identified pollutants are presented with the classification.

The instances of pollution are determined and noted in the system database to enable documentation and tracking. The automated notifications will be created and sent to the administrator whenever a significant pollution was detected to be informed and take timely action. The system can be improved in the future by its modular design that enables it to be integrated with geographic data or real-time monitoring devices. All in all, the methodology guarantees an effective, scalable and systematic nature of assessment of river water quality.

4 ARCHITECTURE DETAILS

A. YOLOv8 Architecture

YOLOv8 will be based on quick and precise object recognition and a simplified and efficient architecture. It adheres to single-stage detection strategy, i.e. object localization and classification is done in a single forward pass of the network. The architecture consists of three components, which are a backbone, a neck, and a head. The backbone derives useful visual characteristics of the input image like edges, shape, and texture. These components are then relayed to the neck which synthesizes information across the different scale of features in enhancing detection of small and large objects in tricky scenes such as river surfaces. The last prediction is done by the head by classifying the objects and drawing bounding boxes on the identified pollutants. This model eliminates anchor boxes and thus, the training process is more flexible and simpler to learn by beginners. It is lightweight and therefore trains and infers faster hence it is capable of real-time environmental monitoring applications where speed and accuracy matter.

B. YOLOv9 Architecture

The YOLOv9 system uses architectural enhancements to boost its ability to learn features and perform object detection. The model is aimed at improved information flow within and across layers, whereby deeper networks learn without significant features being lost. Its backbone is meant to draw more meaningful spatial patterns on the images especially when trying to identify partially visible or overlapping objects on the water environment. The neck component enhances fusion of features in a way that it effectively combines low-level and high-level features, which makes the model to get to know the details of the object, as well as its general context. This leads to higher identification of different contaminants in different lighting and water conditions. The prediction head then takes the combined features and produces the object categories and their locations with a higher degree of stability. The system design achieves high accuracy results while maintaining acceptable performance for computational operations. Because of its systematic construction and better learning ability, this model can be used in complicated detecting assignments where one can observe multiple types of pollution at the same time.

C. YOLOv10 Architecture

YOLOv10 is trained to enhance the performance of the detection through redundancy minimization and the quality of prediction. The architecture is aimed at removing needless parts in the inference process, and thus can lower the computational cost without affecting the accuracy. Its backbone is good at capturing meaningful visual aspects at a minimum amount of noise due to reflection and background components that are common in river images. The neck also improves the feature-matching at varying resolutions, and thus objects are always detected at varying levels. The main characteristic of this system architecture functions through its advanced prediction system which reduces duplicate detection instances while it boosts detection confidence levels. The detection results achieve their most accurate and precise form for documentation and reporting purposes. The system achieves faster processing through its efficient inference pipeline which functions effectively in real-time monitoring system operations. On the whole, this architecture provides a reasonable balance in terms of efficiency, accuracy, and scalability which is why it is appropriate to automated river water quality evaluation.

5 RESULTS AND DISCUSSION

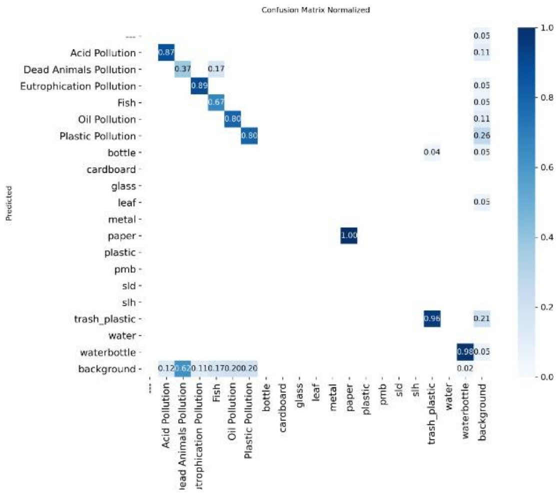

Figure 5 YOLOv8 Confusion Matrix

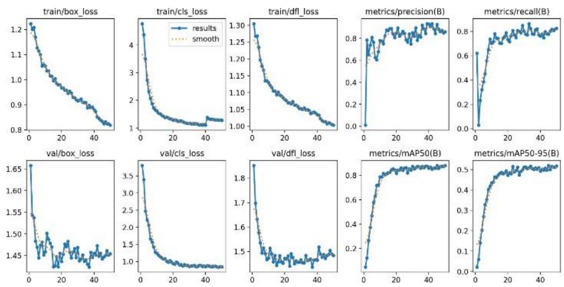

Figure 6 Training And Validation Loss Curves For YOLOV8

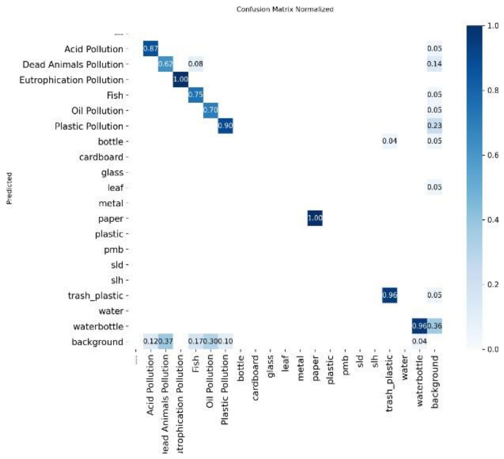

Figure 7 YOLOV10 Confusion Matrix

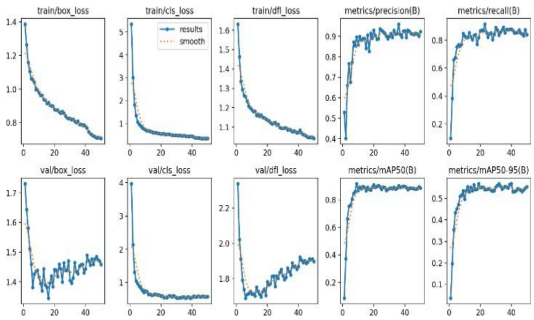

Figure 8 Training And Validation Loss Curves For YOLOV9

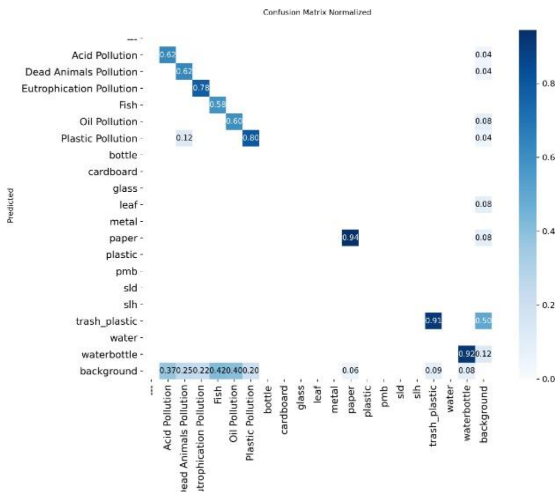

Figure 9 YOLOV10 Confusion Matrix

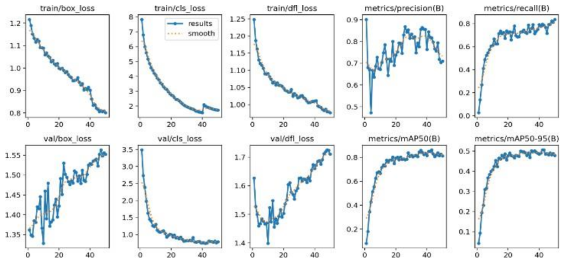

Figure 10 Training And Validation Loss Curves For YOLOV10

Figure 11 Sample Output Images

The normalized confusion matrices were used to evaluate the three detection models and give an idea of the accuracy of each model to detect pollution and the categories of objects. In the confusion matrix of the first model, the object classifications that performed exceptionally well are those of paper, water bottle, and background that scored almost perfect with accuracy score of 1.00 and 0.96, respectively as seen in Figure 5. This implies that the model is capable of detecting frequent debris in the river waters. Nonetheless, misclassifications are great in some classes. As an example, Dead Animals Pollution and Fish are much less accurate with the values of 0.37 and 0.17 accordingly, which means that these two groups are frequently mixed up with the rest of the pollution types or items in the dataset. The performance of Plastic Pollution and Oil Pollution is quite good with 0.86 and 0.80, indicating that the model can identify some of the pollutants in a consistent manner.

In the second model, there is a better detection of certain types of pollution, but it stays very accurate on the objects like metal, trash plastic and water bottle with the accuracy of 1.00 and 0.96 as illustrated in Figure 7. The instances of misclassifications are prominently observed in Dead Animals Pollution, with the lowest figure of correct predictions of 0.08, which may be classified as background. Classification such as Fish and Plastic Pollution are better predicted at 0.90 each and the background is somewhat misleading the model, especially the Acid and oil pollution.

High classification confidence of various objects such as metals, papers and water bottle is also evident in model 3 as shown in Figure 9. But the other two are Dead Animals Pollution and background which is often difficult to predict, often leading to wrong predictions too. In general, these confusion matrices show the strong and weak points of each of the models and which categories can be correctly identified and which need further optimization.

6 CONCLUSION

This project provides a comprehensive design of evaluating the quality of river water basing on automated methods of object detection. With a powerful and easy-to-use web interface and more sophisticated deep learning models, the system can detect various types of pollutants and debris, such as chemical pollutants, plastic waste, dead creatures, and oil stains. The methodology uses image preprocessing, annotation and training of state-of-the-art detection models, and then deploys the models in a modular flask-based platform, which enables uploading of images, automated detection and documentation of the results.

The normalized confusion matrix analysis shows that the models are very accurate with typical debris objects like bottles, paper and metals and some more difficult pollution objects such as dead animals and background because of the similarity in the image and the complicated surroundings. The most recent model of the one that was tested demonstrates an even level of performance regarding detection speed and accuracy in the classification that can be relied on to obtain a good result in a real-time scenario of observing the environment. Further practical value of the system is added by the automated notification and data storage modules which are useful in monitoring the environment and conducting research with the help of the system.

In general, the project demonstrates the opportunities of applying deep learning and interactive platforms to systematically evaluate the river water quality. Not only does the framework decrease the need to have manuals inspected but it also offers an extensible and scalable strategy to have in place as a method of continuous monitoring. Further research in the field could be aimed at better identifying classes of challenging pollution, combining sensor data, and adapting the system to other water bodies, thus playing a role in making the water quality management more holistic and proactive.

7 FUTURE SCOPE

The suggested system can offer a strong structure in terms of river water quality evaluation, and a number of directions can be identified to improve. Among the improvements, it is possible to note the increase in the variety of the dataset in terms of different environmental conditions, seasonal differences, and other types of pollutants. This would assist the detection models to learn more scenarios, which would increase overall accuracy and minimize misclassifications, especially those that are harder to distinguish, like dead animals and background objects.

Connection and integration with real-time sensor networks and Internet of Things (IoT) devices are likely to make the system an entirely automated river monitoring platform. It would be possible to give a complete assessment of the health of rivers in near real-time through the integration of visual detection with chemical and physical water quality measurement.

Furthermore, it could be enhanced by adding explainable artificial intelligence (XAI) techniques so that users and administrators can gain understanding of the logic behind detection results. Another possible direction is the expansion of the system to other water bodies including lakes, reservoirs, and coasts. Lastly, the detection models of the edge devices should be optimized to allow their use in remote sites that require no high computing capabilities, which would facilitate the use of the models in large-scale environmental monitoring and proactive pollution control programs.

8 REFERENCES

A. Heller *et al.*, “Plastic water bottle detection model using computer vision in aquatic environments,” *Sci. Rep.*, vol. 15, no. 1, Dec. 2025, doi: 10.1038/S41598-025-09300-8.

F. A. Khokhar, J. H. Shah, R. Saleem, and A. Masood, “Harnessing deep learning for faster water quality assessment: identifying bacterial contaminants in real time,” *The Visual Computer 2024 41:2*, vol. 41, no. 2, pp. 1037–1048, Apr. 2024, doi: 10.1007/S00371-024-03382-7.

W. Zhu and R. Xu, “Research on an Improved YOLOv8 Algorithm for Water Surface Object Detection,” *Electronics 2025, Vol. 14*, vol. 14, no. 18, Sep. 2025, doi: 10.3390/ELECTRONICS14183615.

N. L. Kirana, D. Kurnianingtyas, and Indriati, “A Deep Learning Approach to Plastic Bottle Waste Detection on the Water Surface using YOLOv6 and YOLOv7,” *Engineering, Technology & Applied Science Research*, vol. 14, no. 6, pp. 18623–18630, Dec. 2024, doi: 10.48084/ETASR.8592.

M. A. Novianta *et al.*, “Monitoring river water quality through predictive modeling using artificial neural networks backpropagation,” *AIMS Environmental Science 2024 4:649*, vol. 11, no. 4, pp. 649–664, 2024, doi:10.3934/ENVIRONSCI.2024032.

Z. Pang, Z. Zhou, J. Fu, W. Jiang, X. Qin, and M. Sun, “Deep learning-based remote sensing retrieval of inland water quality: A review,” *J. Hydrol. Reg. Stud.*, vol. 61, p. 102759, Oct. 2025, doi: 10.1016/J.EJRH.2025.102759.

H. Li *et al.*, “Assessing water quality environmental grades using hyperspectral images and a deep learning model: A case study in Jiangsu, China,” *Ecol. Inform.*, vol. 84, p. 102854, Dec. 2024, doi: 10.1016/J.ECOINF.2024.102854.

J. Mallick, S. Alqadhi, M. Alsubih, M. F. M. Ahmed, and H. G. Abdo, “Automated interpretation of deep learning- based water quality assessment system for enhanced environmental management decisions,” *Applied Water Science 2025 15:5*, vol. 15, no. 5, pp. 110-, Apr. 2025, doi: 10.1007/S13201-025-02452-Y.

L. Kandasamy, A. Mahendran, S. H. V. Sangaraju, P. Mathur, S. V. Faldu, and M. Mazzara, “Enhanced remote sensing and deep learning aided water quality detection in the Ganges River, India supporting monitoring of aquatic environments,” *Results in Engineering*, vol. 25, p. 103604, Mar. 2025, doi: 10.1016/J.RINENG.2024.103604.

S. L. Deshmukh, M. Wilchek, and F. A. Batarseh, “HydroVision: Predicting Optically Active Parameters in Surface Water Using Computer Vision,” Sep. 2025, Accessed: Jan. 27, 2026. [Online]. Available: https://arxiv.org/pdf/2509.01882

K. Vigneswaran, H. Retief, J. C. Holmes, M. G. Andarcia, and H. Tennakoon, “Automated river gauge plate reading using a hybrid object detection and generative AI framework in the Limpopo River Basin,” Dec. 2025, Accessed: Jan. 27, 2026. [Online].Available:https://arxiv.org/pdf/2512.23454

P. S. K, Rathik, S. B. R, and Dr. S. S. V, “A Survey on Real-Time Water Surveillance Boat with Object Detection and Conveyor Belt Retrieval,” *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, vol. 11, no. 6, pp. 185–192, Nov. 2025, doi: 10.32628/CSEIT2511631.

A. Surahmat and D. B. R. W. Yato, “AI-Based Waste Detection for Water Quality Monitoring in the Cisadane River: A Deep Learning Approach,” *Gema Lingkungan Kesehatan*, vol. 23, no. 3, pp. 333–343, Jul. 2025, doi: 10.36568/GELINKES.V23I3.270.

W. Jung, J. Kim, H. Jo, S. Lee, and B. Kim, “Video-Based Deep Learning Approach for Water Level Monitoring in Reservoirs,” *Water 2025, Vol. 17*, vol. 17, no. 17, Aug. 2025, doi: 10.3390/W17172525.

S. Toumi *et al.*, “Harnessing Deep Learning for Real-Time Water Quality Assessment: A Sustainable Solution,” *Water 2024, Vol. 16*, vol. 16, no. 23, Nov. 2024, doi: 10.3390/W16233380.